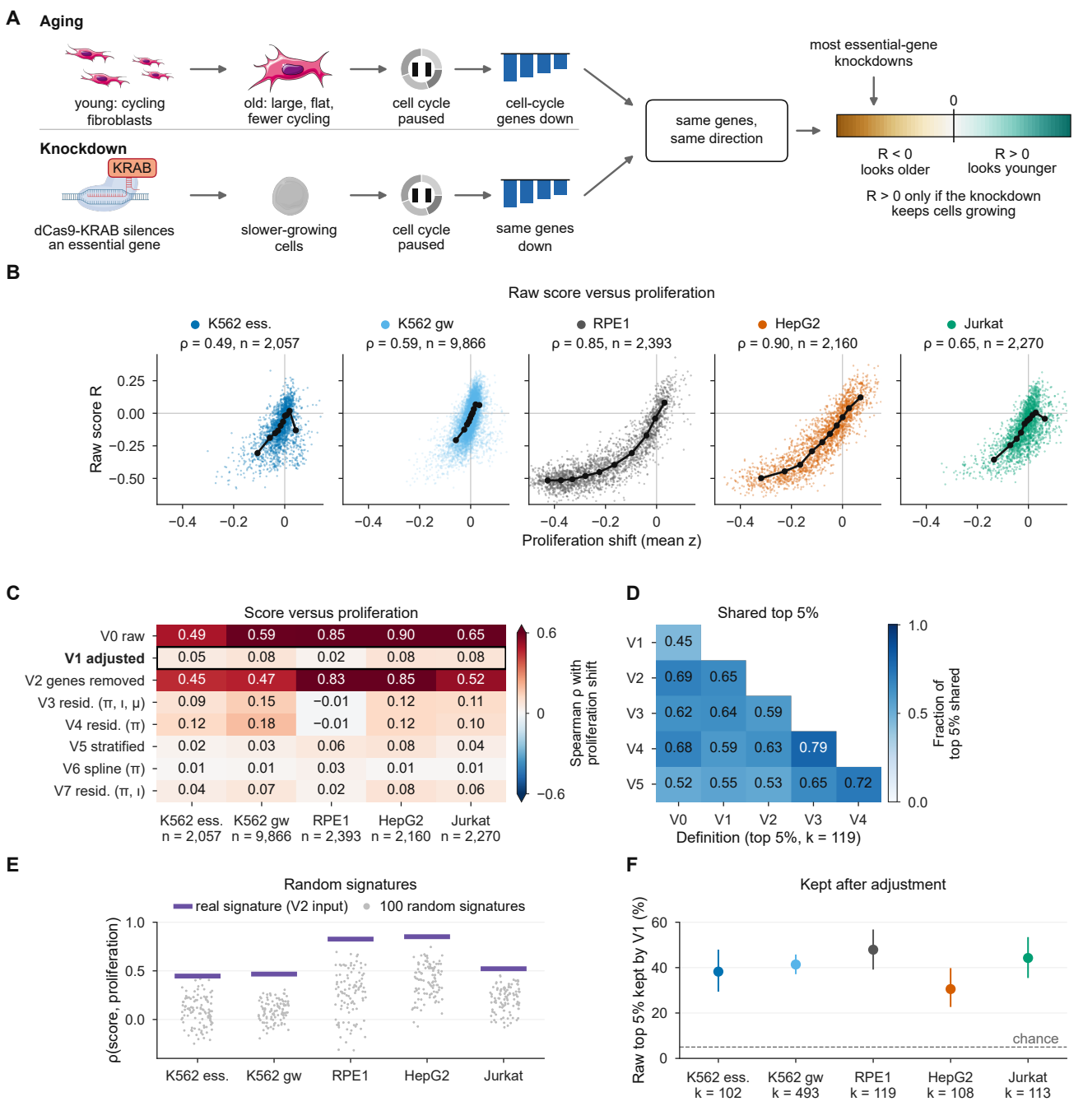

A
Aging
young: cycling fibroblasts
old: large, flat, fewer cycling
cell cycle paused
cell-cycle genes down
Knockdown
KRAB
dCas9-KRAB silences an essential gene
slower-growing cells
cell cycle paused
same genes down
same genes, same direction
most essential-gene knockdowns
0
R < 0 looks older
R > 0 looks younger
R > 0 only if the knockdown keeps cells growing
B
Raw score versus proliferation
K562 ess.
ρ = 0.49, n = 2,057
K562 gw
ρ = 0.59, n = 9,866
RPE1
ρ = 0.85, n = 2,393
HepG2
ρ = 0.90, n = 2,160
Jurkat
ρ = 0.65, n = 2,270
Raw score R
Proliferation shift (mean z)
C
Score versus proliferation
V0 raw 0.49 0.59 0.85 0.90 0.65
V1 adjusted 0.05 0.08 0.02 0.08 0.08
V2 genes removed 0.45 0.47 0.83 0.85 0.52
V3 resid. (π, ι, μ) 0.09 0.15 −0.01 0.12 0.11
V4 resid. (π) 0.12 0.18 −0.01 0.12 0.10
V5 stratified 0.02 0.03 0.06 0.08 0.04
V6 spline (π) 0.01 0.01 0.03 0.01 0.01
V7 resid. (π, ι) 0.04 0.07 0.02 0.08 0.06
K562 ess. n = 2,057
K562 gw n = 9,866
RPE1 n = 2,393
HepG2 n = 2,160
Jurkat n = 2,270
Spearman ρ with proliferation shift
D
Shared top 5%
V1 0.45
V2 0.69 0.65
V3 0.62 0.64 0.59
V4 0.68 0.59 0.63 0.79
V5 0.52 0.55 0.53 0.65 0.72
Definition (top 5%, k = 119)
Fraction of top 5% shared
E
Random signatures
real signature (V2 input)
100 random signatures
ρ(score, proliferation)
F
Kept after adjustment
Raw top 5% kept by V1 (%)
chance
K562 ess. k = 102
K562 gw k = 493
RPE1 k = 119
HepG2 k = 108
Jurkat k = 113